

Numerical Methods

By

Numerical Methods Group

Jacobi's Iteration Method

This method is also known as **the method of simultaneous displacements**.

Consider the system of linear equations:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Here, we assume that the coefficients a_{11} , a_{22} and a_{33} are the largest coefficients in the respective equations so that

$$|a_{11}| > |a_{12}| + |a_{13}|$$

$$|a_{22}| > |a_{21}| + |a_{23}|$$

$$|a_{33}| > |a_{31}| + |a_{32}|$$

Jacobi's iteration method is applicable only if the conditions given above are satisfied

Remark:

If the diagonal element is **equal** to the sum of the absolute values of the other elements in the same row, the Jacobi **method may or may not converge**.

Jacobi's Iteration Method

Now, we can write the system of equations as:

$$x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3]$$

$$x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3]$$

$$x_3 = \frac{1}{a_{33}} [b_3 - a_{31}x_1 - a_{32}x_2]$$

Let the initial approximations be x_1^0, x_2^0 and x_3^0 respectively.

The following iterations are then carried out.

Iteration 1: The first improvements are found as

$$x_{11} = \frac{1}{a_{11}} [b_1 - a_{12}x_2^0 - a_{13}x_3^0]$$

$$x_{21} = \frac{1}{a_{22}} [b_2 - a_{21}x_1^0 - a_{23}x_3^0]$$

$$x_{31} = \frac{1}{a_{33}} [b_3 - a_{31}x_1^0 - a_{32}x_2^0]$$

Jacobi's Iteration Method

Iteration 2: The second improvements are obtained as

$$x_{12} = \frac{1}{a_{11}} [b_1 - a_{12}x_{21} - a_{13}x_{31}]$$

$$x_{22} = \frac{1}{a_{22}} [b_2 - a_{21}x_{11} - a_{23}x_{31}]$$

$$x_{32} = \frac{1}{a_{33}} [b_3 - a_{31}x_{11} - a_{32}x_{21}]$$

The above iteration process is continued until the values of x_1 , x_2 and x_3 are found to a pre-assigned degree of accuracy. That is, the procedure is continued until the relative error between two consecutive vector norm is satisfactorily small.

In Jacobi's method, it is a general practice to assume

$$x_1^0 = x_2^0 = x_3^0 = 0$$

The method can be extended to a system of n linear simultaneous equations in n unknowns.

Example :

Solve the following equations by Jacobi's method:

$$15x + 3y - 2z = 85$$

$$2x + 10y + z = 51$$

$$x - 2y + 8z = 5$$

Solution :

In the above equations:

$$|15| > |3| + |-2|$$

$$|10| > |2| + |1|$$

$$|8| > |1| + |-2|$$

then Jacobi's method is applicable.

Now, we can write the system of equations as:

$$x = \frac{1}{15} [85 - 3y + 2z]$$

$$y = \frac{1}{10} [51 - 2x - z]$$

$$z = \frac{1}{8} [5 - x + 2y]$$

Let the initial approximations be $x^0 = y^0 = z^0 = 0$.

Iteration 1: The first improvements are found as

$$x_1 = \frac{1}{15} [85 - 3y^0 + 2z^0] = \frac{85}{15}$$

$$y_1 = \frac{1}{10} [51 - 2x^0 - z^0] = \frac{51}{10}$$

$$z_1 = \frac{1}{8} [5 - x^0 + 2y^0] = \frac{5}{8}$$

Iteration 2: The second improvements are obtained as

$$x_2 = \frac{1}{15} [85 - 3y_1 + 2z_1] = \frac{1}{15} \left[85 - 3 \frac{51}{10} + 2 \frac{5}{8} \right] = 4.73$$

$$y_2 = \frac{1}{10} [51 - 2x_1 - z_1] = \frac{1}{10} \left[51 - 2 \frac{85}{15} - \frac{5}{8} \right] = 3.904$$

$$z_2 = \frac{1}{8} [5 - x_1 + 2y_1] = \frac{1}{8} \left[5 - \frac{85}{15} + 2 \frac{51}{10} \right] = 1.192$$

Iteration 3: The third improvements are obtained as

$$x_3 = \frac{1}{15} [85 - 3y_2 + 2z_2] = \frac{1}{15} [85 - 3(3.904) + 2(1.192)] = 5.045$$

$$y_3 = \frac{1}{10} [51 - 2x_2 - z_2] = \frac{1}{10} [51 - 2(4.73) - (1.192)] = 4.035$$

$$z_3 = \frac{1}{8} [5 - x_2 + 2y_2] = \frac{1}{8} [5 - (4.73) + 2(3.904)] = 1.010$$

Iteration 4:

$$x_4 = \frac{1}{15} [85 - 3y_3 + 2z_3] = \frac{1}{15} [85 - 3(4.035) + 2(1.010)] = 4.994$$

$$y_4 = \frac{1}{10} [51 - 2x_3 - z_3] = \frac{1}{10} [51 - 2(5.045) - (1.010)] = 3.99$$

$$z_4 = \frac{1}{8} [5 - x_3 + 2y_3] = \frac{1}{8} [5 - (5.045) + 2(4.035)] = 1.003$$

Iteration 5:

$$x_5 = \frac{1}{15} [85 - 3y_4 + 2z_4] = \frac{1}{15} [85 - 3(3.99) + 2(1.003)] = 5.002$$

$$y_5 = \frac{1}{10} [51 - 2x_4 - z_4] = \frac{1}{10} [51 - 2(4.994) - (1.003)] = 4.001$$

$$z_5 = \frac{1}{8} [5 - x_4 + 2y_4] = \frac{1}{8} [5 - (4.994) + 2(3.99)] = 0.998$$

Iteration 6:

$$x_6 = \frac{1}{15} [85 - 3y_5 + 2z_5] = \frac{1}{15} [85 - 3(4.001) + 2(0.998)] = 5.0$$

$$y_6 = \frac{1}{10} [51 - 2x_5 - z_5] = \frac{1}{10} [51 - 2(5.002) - (0.998)] = 4.0$$

$$z_6 = \frac{1}{8} [5 - x_5 + 2y_5] = \frac{1}{8} [5 - (5.002) + 2(4.001)] = 1.0$$

Iteration 7:

$$x_7 = \frac{1}{15} [85 - 3y_6 + 2z_6] = \frac{1}{15} [85 - 3(4) + 2(1)] = 5.0$$

$$y_7 = \frac{1}{10} [51 - 2x_6 - z_6] = \frac{1}{10} [51 - 2(5) - (1)] = 4.0$$

$$z_7 = \frac{1}{8} [5 - x_6 + 2y_6] = \frac{1}{8} [5 - (5) + 2(4)] = 1.0$$

Example :

Use Jacobi's method to obtain the solutions of the system of equations correct to three decimal places:

$$x + 2y + z = 0$$

$$3x + y - z = 0$$

$$x - y + 4z = 3$$

Solution :

Rearrange the equations in such a way that all the diagonal terms are dominant. (العنصر القطري في كل صف أكبر من مجموع القيم المطلقة لباقي عناصر الصف).

$$3x + y - z = 0$$

$$x + 2y + z = 0$$

$$x - y + 4z = 3$$

In the above equations:

$$|3| > |1| + |-1|$$

$$|2| > |1| + |1|$$

$$|4| > |1| + |-1|$$

then Jacobi's method is applicable.

Now, we can write the system of equations as:

$$x = \frac{1}{3}[-y + z]$$

$$y = \frac{1}{2}[-x - z]$$

$$z = \frac{1}{4}[3 - x + y]$$

Let the initial approximations be $x^0 = y^0 = z^0 = 0$.

Iteration 1: The first improvements are found as

$$x_1 = \frac{1}{3}[-y^0 + z^0] = 0$$

$$y_1 = \frac{1}{2}[-x^0 - z^0] = 0$$

$$z_1 = \frac{1}{4}[3 - x^0 + y^0] = \frac{3}{4}$$

Iteration 2: The second improvements are obtained as

$$x_2 = \frac{1}{3}[-y_1 + z_1] = \frac{1}{3}\left[0 + \frac{3}{4}\right] = 0.25$$

$$y_2 = \frac{1}{2}[-x_1 - z_1] = \frac{1}{2}\left[0 - \frac{3}{4}\right] = -0.375$$

$$z_2 = \frac{1}{4}[3 - x_1 + y_1] = \frac{1}{4}[3 - 0 + 0] = 0.75$$

Iteration 3: The third improvements are obtained as

$$x_3 = \frac{1}{3}[-y_2 + z_2] = \frac{1}{3}[0.375 + 0.75] = 0.375$$

$$y_3 = \frac{1}{2}[-x_2 - z_2] = \frac{1}{2}[-0.25 - 0.75] = -0.5$$

$$z_3 = \frac{1}{4}[3 - x_2 + y_2] = \frac{1}{4}[3 - (0.25) + (-0.375)] = 0.594$$

Iteration 4:

$$x_4 = \frac{1}{3}[-y_3 + z_3] = \frac{1}{3}[0.5 + 0.594] = 0.3647$$

$$y_4 = \frac{1}{2}[-x_3 - z_3] = \frac{1}{2}[-0.375 - 0.594] = -0.4845$$

$$z_4 = \frac{1}{4}[3 - x_3 + y_3] = \frac{1}{4}[3 - (0.375) + (-0.5)] = 0.5313$$

Iteration 5:

$$x_5 = \frac{1}{3}[-y_4 + z_4] = \frac{1}{3}[0.4845 + 0.5313] = 0.3386$$

$$y_5 = \frac{1}{2}[-x_4 - z_4] = \frac{1}{2}[-0.3647 - 0.5313] = -0.448$$

$$z_5 = \frac{1}{4}[3 - x_4 + y_4] = \frac{1}{4}[3 - (0.3647) + (-0.4845)] = 0.5377$$

Iteration 6:

$$x_6 = \frac{1}{3}[-y_5 + z_5] = \frac{1}{3}[0.448 + 0.5377] = 0.3286$$

$$y_6 = \frac{1}{2}[-x_5 - z_5] = \frac{1}{2}[-0.3386 - 0.5377] = -0.438$$

$$z_6 = \frac{1}{4}[3 - x_5 + y_5] = \frac{1}{4}[3 - (0.3386) + (-0.448)] = 0.5534$$

Iteration 7:

$$x_7 = \frac{1}{3}[-y_6 + z_6] = \frac{1}{3}[0.438 + 0.5534] = 0.3305$$

$$y_7 = \frac{1}{2}[-x_6 - z_6] = \frac{1}{2}[-0.3286 - 0.5534] = -0.441$$

$$z_7 = \frac{1}{4}[3 - x_6 + y_6] = \frac{1}{4}[3 - (0.3286) + (-0.438)] = 0.5584$$

Iteration 8:

$$x_8 = \frac{1}{3}[-y_7 + z_7] = \frac{1}{3}[0.441 + 0.5584] = 0.3331$$

$$y_8 = \frac{1}{2}[-x_7 - z_7] = \frac{1}{2}[-0.3305 - 0.5584] = -0.44445$$

$$z_8 = \frac{1}{4}[3 - x_7 + y_7] = \frac{1}{4}[3 - (0.3305) + (-0.441)] = 0.5571$$

Iteration 9:

$$x_9 = \frac{1}{3}[-y_8 + z_8] = \frac{1}{3}[0.44445 + 0.5571] = 0.33385$$

$$y_9 = \frac{1}{2}[-x_8 - z_8] = \frac{1}{2}[-0.3331 - 0.5571] = -0.4451$$

$$z_9 = \frac{1}{4}[3 - x_8 + y_8] = \frac{1}{4}[3 - (0.3331) + (-0.44445)] = 0.5556$$

Iteration 10:

$$x_{10} = \frac{1}{3}[-y_9 + z_9] = \frac{1}{3}[0.4451 + 0.5556] = 0.33357$$

$$y_{10} = \frac{1}{2}[-x_9 - z_9] = \frac{1}{2}[-0.33385 - 0.5556] = -0.4447$$

$$z_{10} = \frac{1}{4}[3 - x_9 + y_9] = \frac{1}{4}[3 - (0.33385) + (-0.4451)] = 0.5553$$

so to three decimal places the approximate solution

$$x = 0.333, \quad y = -0.444, \quad z = 0.555$$

Gauss-Seidal Iteration Method

This method is also known as **the method of successive approximations**.

Consider the system of linear equations:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Here, we assume that the coefficients a_{11} , a_{22} and a_{33} are the largest coefficients in the respective equations so that

$$|a_{11}| \geq |a_{12}| + |a_{13}|$$

$$|a_{22}| \geq |a_{21}| + |a_{23}|$$

$$|a_{33}| \geq |a_{31}| + |a_{32}|$$

Gauss-Seidal iteration method is applicable only if the conditions given above are satisfied

Gauss-Seidal Iteration Method

Now, we can write the system of equations as:

$$x_1 = \frac{1}{a_{11}} [b_1 - a_{12}x_2 - a_{13}x_3]$$

$$x_2 = \frac{1}{a_{22}} [b_2 - a_{21}x_1 - a_{23}x_3]$$

$$x_3 = \frac{1}{a_{33}} [b_3 - a_{31}x_1 - a_{32}x_2]$$

Let the initial approximations be x_1^0, x_2^0 and x_3^0 respectively.

The following iterations are then carried out.

Iteration 1: The first improvements are found as

$$x_{11} = \frac{1}{a_{11}} [b_1 - a_{12}x_2^0 - a_{13}x_3^0]$$

$$x_{21} = \frac{1}{a_{22}} [b_2 - a_{21}x_{11} - a_{23}x_3^0]$$

$$x_{31} = \frac{1}{a_{33}} [b_3 - a_{31}x_{11} - a_{32}x_{21}]$$

Gauss-Seidal Iteration Method

Iteration 2: The second improvements are obtained as

$$x_{12} = \frac{1}{a_{11}} [b_1 - a_{12}x_{11} - a_{13}x_{31}]$$

$$x_{22} = \frac{1}{a_{22}} [b_2 - a_{21}x_{12} - a_{23}x_{31}]$$

$$x_{32} = \frac{1}{a_{33}} [b_3 - a_{31}x_{12} - a_{32}x_{22}]$$

The above iteration process is continued until the values of x_1 , x_2 and x_3 are found to a pre-assigned degree of accuracy. That is, the procedure is continued until the relative error between two consecutive vector norm is satisfactorily small.

In Gauss-Seidal iteration method, it is a general practice to assume

$$x_1^0 = x_2^0 = x_3^0 = 0$$

The method can be extended to a system of n linear simultaneous equations in n unknowns.

Example :

Solve the following equations by Gauss-Seidal method:

$$8x + 2y - 2z = 8$$

$$x - 8y + 3z = -4$$

$$2x + y + 9z = 12$$

Solution :

In the above equations:

$$|8| \geq |2| + |-2|$$

$$|-8| \geq |1| + |3|$$

$$|9| \geq |2| + |1|$$

then Gauss-Seidal method is applicable.

Now, we can write the system of equations as:

$$\begin{aligned}x &= \frac{1}{8}[8 - 2y + 2z] \\y &= \frac{-1}{8}[-4 - x - 3z] \\z &= \frac{1}{9}[12 - 2x - y]\end{aligned}$$

Let the initial approximations be $x^0 = y^0 = z^0 = 0$.

Iteration 1: The first improvements are found as

$$\begin{aligned}x_1 &= \frac{1}{8}[8 - 2y^0 + 2z^0] = 1 \\y_1 &= \frac{-1}{8}[-4 - x_1 - 3z^0] = 0.625 \\z_1 &= \frac{1}{9}[12 - 2x_1 - y_1] = 1.042\end{aligned}$$

Iteration 2: The second improvements are obtained as

$$x_2 = \frac{1}{8} [8 - 2y_1 + 2z_1] = 1.104$$

$$y_2 = \frac{-1}{8} [-4 - x_2 - 3z_1] = 1.029$$

$$z_2 = \frac{1}{9} [12 - 2x_2 - y_2] = 0.974$$

Iteration 3: The third improvements are obtained as

$$x_3 = \frac{1}{8} [8 - 2y_2 + 2z_2] = 0.986$$

$$y_3 = \frac{-1}{8} [-4 - x_3 - 3z_2] = 0.989$$

$$z_3 = \frac{1}{9} [12 - 2x_3 - y_3] = 1.004$$

Iteration 4:

$$x_4 = \frac{1}{8}[8 - 2y_3 + 2z_3] = 1.004$$

$$y_4 = \frac{-1}{8}[-4 - x_4 - 3z_3] = 1.002$$

$$z_4 = \frac{1}{9}[12 - 2x_4 - y_4] = 0.999$$

Iteration 5:

$$x_5 = \frac{1}{8}[8 - 2y_4 + 2z_4] = 0.999$$

$$y_5 = \frac{-1}{8}[-4 - x_5 - 3z_4] = 1.0$$

$$z_5 = \frac{1}{9}[12 - 2x_5 - y_5] = 1.0$$

Iteration 6:

$$x_6 = \frac{1}{8} [8 - 2y_5 + 2z_5] = 1.0$$

$$y_6 = \frac{-1}{8} [-4 - x_6 - 3z_5] = 1.0$$

$$z_6 = \frac{1}{9} [12 - 2x_6 - y_6] = 1.0$$

Hence, the solution is given by $x = 1, y = 1$ and $z = 1$.

Example :

Solve the following equations by Gauss-Seidal method:

$$x + 2y + z = 0$$

$$3x + y - z = 0$$

$$x - y + 4z = 3$$

Solution :

$$3x + y - z = 0$$

$$x + 2y + z = 0$$

$$x - y + 4z = 3$$

In the above equations:

$$|3| > |1| + |-1|$$

$$|2| > |1| + |1|$$

$$|4| > |1| + |-1|$$

then Gauss-Seidal method is applicable.

Now, we can write the system of equations as:

$$x = \frac{1}{3}[-y + z]$$

$$y = \frac{1}{2}[-x - z]$$

$$z = \frac{1}{4}[3 - x + y]$$

Let the initial approximations be $x^0 = y^0 = z^0 = 0$.

Iteration 1: The first improvements are found as

$$x_1 = \frac{1}{3}[-y^0 + z^0] = 0$$

$$y_1 = \frac{1}{2}[-x_1 - z^0] = 0$$

$$z_1 = \frac{1}{4}[3 - x_1 + y_1] = 0.75$$

Iteration 2: The second improvements are obtained as

$$x_2 = \frac{1}{3}[-y_1 + z_1] = 0.25$$

$$y_2 = \frac{1}{2}[-x_2 - z_1] = -0.5$$

$$z_2 = \frac{1}{4}[3 - x_2 + y_2] = 0.5625$$

Iteration 3: The third improvements are obtained as

$$x_3 = \frac{1}{3}[-y_2 + z_2] = 0.354$$

$$y_3 = \frac{1}{2}[-x_3 - z_2] = -0.458$$

$$z_3 = \frac{1}{4}[3 - x_3 + y_3] = 0.547$$

Iteration 4:

$$x_4 = \frac{1}{3}[-y_3 + z_3] = 0.335$$

$$y_4 = \frac{1}{2}[-x_4 - z_3] = -0.441$$

$$z_4 = \frac{1}{4}[3 - x_4 + y_4] = 0.556$$

Iteration 5:

$$x_5 = \frac{1}{3}[-y_4 + z_4] = 0.3323$$

$$y_5 = \frac{1}{2}[-x_5 - z_4] = -0.4441$$

$$z_5 = \frac{1}{4}[3 - x_5 + y_5] = 0.5559$$

Iteration 6:

$$x_6 = \frac{1}{3}[-y_5 + z_5] = 0.33333$$

$$y_6 = \frac{1}{2}[-x_6 - z_5] = -0.4446$$

$$z_6 = \frac{1}{4}[3 - x_6 + y_6] = 0.55552$$

Hence, the solution is given by $x = 0.333$, $y = -0.444$ and $z = 0.555$.

Example :

Solve the following equations by Gauss-Seidal method:

$$\begin{aligned}4x - y + z &= 12 \\-x + 4y - 2z &= -1 \\x - 2y + 4z &= 5\end{aligned}$$

Solution :

Try by yourself

**THANK YOU
FOR YOUR
ATTENTION**

